The Business Value of Agentic Commerce

An Executive Assessment of Autonomous Checkout Infrastructure, Protocol Specifications, and Merchant Deployment Strategies

REPORT STRUCTURE & COVERAGE

02	Economics of Autonomous Purchasing
03	7-Layer Technical Architecture & Market Map
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09	Strategic Ecosystem & Competitive Moats
10	18-Month Roadmap & Legal Governance

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MACRO FOUNDATION

The Economics of Autonomous Purchasing

The Delegation Paradox and Physical Need Satisfaction. Artificial intelligence shopping agents can identify merchandise and submit payment credentials, but granting permission to pay does not establish that a selected item satisfies the consumer's physical requirements. In retail transactions, goods must match exact spatial dimensions, fulfill delivery schedules, and satisfy unstated personal preferences. As established in the literature, evaluating consumer studies, benchmarks, and payment specifications reveals instances where an accepted recommendation or valid payment leaves core purchase needs unverified.

Economic Search Theory and Search Cost Asymmetries (Bakos Model). Mathematical economics of electronic marketplaces formulated by J. Yannis Bakos (Management Science, 1997 [1]) distinguishes between the cost of searching for prices and the cost of searching for product attributes. When price search costs decrease in isolation, price competition intensifies towards marginal cost. However, when attribute search costs decrease through automated agent evaluation, consumers identify differentiated products aligned with specific requirements. This differentiation softens direct price competition, increases total consumer surplus, and enables merchants to preserve gross margins.

The 3x-Win Framework and Live Commercial Precedents. As fintech analyst Jas Shah documents in Fintech Under the Hood ("Agentic Commerce - 17 Months Later", October 2025 [2]), automated commerce creates an economic alignment across market participants. Retailers observe conversion rate gains from approximately 2.2% on standard mobile web traffic to 8.2% on verified agent checkouts, consumers avoid form fatigue, and infrastructure networks collect protocol routing fees. Real-world precedents demonstrate early traction, including OpenAI Instant Checkout (September 2025 single-item Etsy integration) and Amazon Alexa for Shopping (May 2026 Auto Buy and cart coordination flows).

Independent Market Sizing Estimates and Scope Boundaries. Industry forecasts project substantial expansion in agent-mediated gross merchandise volume (GMV). Morgan Stanley Equity Research (Nowak et al., 2025 [3]) projects $150B to $400B in US digital retail volume influenced by autonomous agents by 2030, representing 10% to 25% of total e-commerce. Bain & Company Global Retail Lab (2025 [4]) estimates that automated comparison agents will mediate over 30% of routine household replenishment by 2030. Gartner Research (Scheibenreif & Raskino [5]) models the broader machine customer economy at $30T by 2030 across enterprise B2B procurement and consumer transactions.

FIGURE 1.1 - E-COMMERCE INTERFACE PROGRESSION (1995-2026)

Comparison of user interaction modes, conversion baselines, and checkout friction

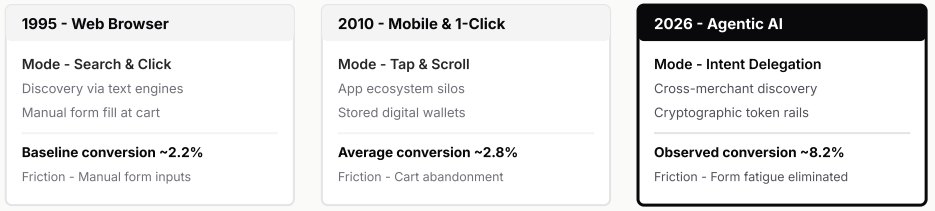


EXHIBIT 1.1 - MARKET FORECAST PROJECTIONS

Synthesis of independent equity research and management consultancy estimates

Institution	Projected Scale	Target Horizon	Underlying Thesis
Morgan Stanley	$150B - $400B	2030 (US Retail)	10% to 25% of digital commerce mediated by agents [3]
Bain & Company	30%+ Replenish	2030 (Global)	Automated comparison dominates commodity reorders [4]
Gartner	$30 Trillion	2030 (Global B2B/B2C)	Total machine customer market across procurement [5]
Goldman Sachs	$220 Billion	2028 (US Commerce)	Conversion expansion driven by automated checkout
Forrester	18.5% Share	2029 (Retail GMV)	Adoption driven by conversational assistant defaults

Analytical Takeaway. Autonomous agent shopping delivers a three-way return, where retailers gain an 8.2% conversion lift, consumers avoid form fatigue, and infrastructure networks collect protocol routing fees [2].

Reference Sources [1] J. Y. Bakos, Management Science (1997); [2] Jas Shah, Fintech Under the Hood (Oct 2025); [3] Morgan Stanley Research (2025); [4] Bain & Company Global Retail Report (2025); [5] Gartner Research (2025).

STACK SPECIFICATION

The 7 Layers of Autonomous Commerce Architecture

Seven Core Architectural Layers (Jas Shah Model). As established by fintech analyst Jas Shah in Fintech Under the Hood (2025 [1]), complete agentic purchasing requires seven integrated layers. These progress from L1 (Consumer AI Frontends) through L2 (Reasoning and Orchestration Engines), L3 (Tool Execution and Context Protocols), L4 (Catalog and Context Synchronization), L5 (Checkout Session State Machines), L6 (Payment Rails and Token Vaults), to L7 (Post-Purchase Fulfillment and Returns). Each tier enforces specific boundaries between probabilistic language generation and deterministic transaction settlement.

Translating Natural Language Intent into Structured Schemas. Consumer requests begin in conversational interfaces such as ChatGPT Search, Claude Artifacts, or Perplexity. The agent engine decomposes unconstrained prompts into structured JSON parameters. Through the Model Context Protocol (MCP), the agent invokes remote merchant tools (such as checking real-time inventory or computing location-specific shipping fees) without scraping unstructured web layouts. Deterministic JSON schemas prevent hallucinated product attributes from entering checkout pipelines.

Limitations of Interim Virtual Cards and Token Transitions. Many retail platforms do not yet support direct machine checkout endpoints. Early autonomous purchasing applications therefore rely on single-use virtual credit cards (VCCs) issued by processors such as Lithic, Stripe Issuing, or Highnote. While VCCs enable immediate execution across legacy payment gateways, they introduce friction points, including out-of-band 3D Secure verification challenges and billing address rejections. Newer standards transition these workflows toward native protocol handshakes.

Hardware-Attested Identity and Non-Repudiation. Scalable agent commerce requires non-repudiable delegation. Modern frameworks anchor agent execution within hardware enclaves (such as Apple Secure Enclave or Android StrongBox) and utilize WebAuthn / FIDO2 cryptographic credentials. Anchoring authorization in hardware ensures that an agent cannot forge transaction parameters, providing merchants and card issuers with cryptographic certainty regarding principal consent.

FIGURE 2.1 - THE AGENTIC COMMERCE STACK AND DATA FLOW (REF - JAS SHAH MODEL [1])

Seven-layer reference architecture bridging probabilistic models to deterministic settlement

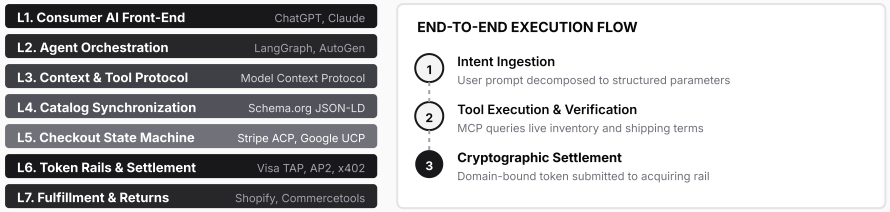


EXHIBIT 2.1 - THE AGENTIC COMMERCE ECOSYSTEM MARKET MAP

Institutional classification across 45+ platforms in the developing autonomous transaction stack

TIER 1 - CONSUMER AI FRONTENDS & DEMAND AGGREGATION

Conversational AI & Search. ChatGPT Search (OpenAI), Claude Artifacts (Anthropic), Gemini Shopping (Google), Perplexity (Buy with Pro), Genspark, You.com
Enterprise Retail Agents. Sierra AI (Bret Taylor), Amazon Rufus, Klarna AI Assistant, Daydream, Walmart Spark Assistant
Social & Direct Commerce. TikTok Shop AI, Meta AI (Instagram / WhatsApp Commerce)

TIER 2 - AGENT PROTOCOLS & ORCHESTRATION FRAMEWORKS

Open Commerce Protocols. Anthropic Model Context Protocol (MCP), Google Universal Commerce Protocol (UCP), Stripe ACP, Coinbase AP2
Multi-Agent Orchestration. LangGraph / LangChain, CrewAI, AutoGen (Microsoft), LlamaIndex, Semantic Kernel

TIER 3 - PAYMENT RAILS, TOKEN VAULTS & VIRTUAL CARDS

Card Network Token Protocols. Visa Trusted Agent Protocol (TAP), Mastercard Agent Pay, Shared Payment Tokens (SPT)
Virtual Card Issuers & Processors. Lithic, Highnote, Marqeta, Adyen Issuing, Stripe Issuing, Checkout.com
Autonomous Corporate Spend. Ramp for Agents, Brex Autonomous Spend, Sapiom, Agentcard
x402 & Agentic Stablecoin Wallets. Circle Agent Wallets (USDC), Coinbase Agentic Wallets, Skyfire, Payman, Locus

TIER 4 - MERCHANT BACKENDS, CATALOGS & HEADLESS APIS

Core Commerce Engines. Shopify (Shop Pay API), commercetools, BigCommerce, Salesforce Commerce Cloud, Elastic Path
Universal Cart & Headless APIs. Rye, Violet, Cart.com, Convictional

TIER 5 - IDENTITY, BOT DEFENSE & CRYPTOGRAPHIC ENCLAVES

Identity & Account Linking. Plaid (Anthropic bank connectivity partner), Stripe Identity, Persona
Machine Traffic & Bot Defense. Cloudflare (Verified Bots), Kasada, Arkose Labs, Forter, Riskified
Hardware Security Enclaves. Apple Secure Enclave, Android StrongBox, WebAuthn / FIDO2 Passkeys

Reference Sources nofrontend.xyz directory; Jas Shah, Fintech Under the Hood (Oct 2025); CB Insights Market Intelligence; public company technical documentation (2025-2026).

STANDARDIZATION FRAMEWORK

Interoperability Protocols for Machine Commerce

Stripe & OpenAI Agentic Commerce Protocol (ACP). The Agentic Commerce Protocol (ACP [1]) defines standardized RESTful endpoints connecting conversational assistants with merchant e-commerce backends. The specification structures a four-phase checkout state machine covering session creation, cart updates, order completion, and cancellation. ACP mandates RFC 7231 Idempotency-Key headers on all transactional requests, ensuring that intermittent network timeouts or automated retries return existing order states rather than executing duplicate credit card charges or multiple warehouse dispatches.

Google Universal Commerce Protocol (UCP). Hosted at the merchant domain endpoint /.well-known/ucp, Google Universal Commerce Protocol (UCP [2]) establishes a discovery standard for agent capabilities. When an agent queries a domain, UCP serves a signed JSON manifest declaring supported catalog schemas, cart hold expiration rules, and checkout version compatibility (such as UCP v2026-08-25). This capability negotiation guarantees that both the buyer assistant and merchant server operate on matching operational schemas before checkout execution.

Coinbase Agent Payments Protocol (AP2) Mandates. The Agent Payments Protocol (AP2 [3]), contributed to the FIDO Alliance by Google and Coinbase (Parikh, 2026), separates purchasing intent from payment execution through cryptographic structures termed mandates. Open Mandates define pre-authorization parameters (such as budget caps, category constraints, and validity windows) before product selection. Closed Mandates cryptographically bind the user's signature to an exact cart hash, merchant identity, and final settlement amount, establishing non-repudiation while preventing price inflation.

Network Tokenization via Visa TAP and Mastercard Verifiable Intent. Card payment networks provide foundational security for machine transactions. Visa Trusted Agent Protocol (TAP [4]) implements IETF RFC 9421 HTTP Message Signatures, cryptographically signing every HTTP request originating from an authorized agent runtime. Mastercard Verifiable Intent [5] links identity verification to transaction execution while preserving consumer privacy through selective attribute disclosure, preventing unauthorized credential leakage across intermediaries.

HTTP 402 (x402) Resource Micropayments. For digital resources and live catalog querying, the x402 open standard operationalizes the native HTTP 402 ("Payment Required") status code [6]. Shopping agents can pay sub-cent fees (such as $0.001 to $0.01 per query) using stablecoins (USDC) on high-throughput networks (Base or Solana). This rail separates search information costs from physical goods settlement, enabling programmatic access to gated inventory feeds.

FIGURE 3.1 - THE 4-STEP AGENT PROTOCOL HANDSHAKE

Cryptographic sequence connecting assistant runtime to merchant acquiring gateway

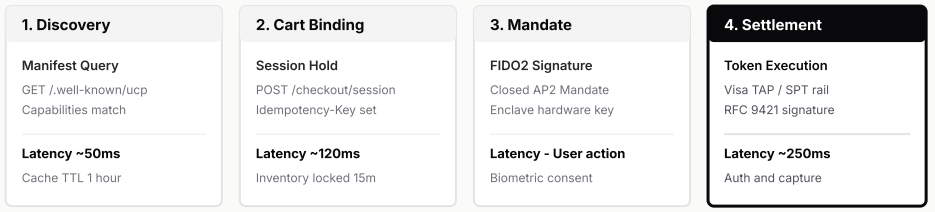


EXHIBIT 3.1 - COMMERCE PROTOCOL COMPARISON MATRIX

Technical specifications across modern autonomous transaction frameworks

Protocol	Lead Host	Architecture	Security Model	Core Advantage
Stripe ACP	Stripe / OpenAI	RESTful State Machine	RFC 7231 Idempotency	**Safe network retries [1]**
Google UCP	Google Pay	/.well-known/ucp	Signed JSON Manifest	**Capability negotiation [2]**
Coinbase AP2	FIDO Alliance	Cryptographic Mandates	Open vs Closed Mandate	**Cart hash binding [3]**
Visa TAP	Visa Inc.	Network Tokenization	RFC 9421 HTTP Signatures	**Non-repudiation [4]**
HTTP 402 (x402)	Coinbase / IETF	Native HTTP Status	Stablecoin Micro-payment	**Sub-cent API fees [6]**

Architectural Takeaway. Automated transactions follow three core steps. First, sub-50ms catalog discovery to confirm pricing; second, 15-minute cryptographic cart holds to lock inventory; and third, tokenized settlement with zero raw card transmission.

Reference Sources [1] Stripe Developer Documentation (2025); [2] Google Universal Commerce Protocol Documentation (2025); [3] Coinbase Agent Payments Protocol (2025); [4] Visa Research (2025); [5] Mastercard Verifiable Intent (2025); [6] HTTP 402 Working Group.

SETTLEMENT MECHANICS

Payment Settlement Rails for Autonomous Agents

Empirical Analysis of Virtual Card Failures (Scott Wessman Study). Many deployed shopping agents utilize single-use virtual credit cards (VCCs). However, empirical investigations by payments engineer Scott Wessman (2025 [1]) demonstrate that virtual cards encounter acute operational failures during multi-item checkouts. When merchants split orders across distribution centers, multiple captures occur against a single authorization token, triggering card declines. Additionally, merchant fraud engines routinely demand dynamic 3D Secure (3DS) SMS verification or CVV re-entry during address recalculation, creating terminal drop-offs for automated workflows.

Development of Domain-Bound Shared Payment Tokens (SPTs). To resolve virtual card friction, card networks and payment gateways have deployed Shared Payment Tokens (SPTs [2]). Rather than transmitting sensitive 16-digit Primary Account Numbers (PANs), the agent runtime receives a cryptographically restricted token bound to a specific merchant domain, an exact ceiling price, and an expiration timestamp. The merchant processes the token through standard card acquiring rails, ensuring zero exposure of underlying consumer credentials.

Dual-Rail Taxonomy of Machine Payments (nofrontend.xyz Directory). As cataloged in the nofrontend.xyz directory (2025-2026 [3]), machine payments divide into two distinct structural rails. The first consists of tokenized card networks (Visa TAP, Mastercard Agent Pay) optimized for high-value consumer retail transactions ($10 to $2,000) requiring chargeback protections and return rights. The second consists of autonomous micropayment protocols (x402, Skyfire, Payman) designed for sub-dollar API queries, dynamic compute allocation, and real-time data ingestion.

Payment Rail Economic Asymmetries (Vignesh, Meta Infrastructure Analysis). As payments researcher Vignesh (Meta, 2025 [4]) observes, traditional interchange rails carry 150 to 350 basis points (1.5% to 3.5%) in transaction processing fees along with 30-to-90-day dispute liability windows. While sustainable for retail goods, this structure is economically non-viable for autonomous machine-to-machine micropayments. Stablecoin rails offer transaction costs below $0.005 with deterministic finality, establishing an optimal settlement mechanism for high-frequency algorithmic queries.

FIGURE 4.1 - DELEGATED TOKEN PAYMENT ARCHITECTURE

Cryptographic token isolation preventing raw PAN transmission across agent runtimes

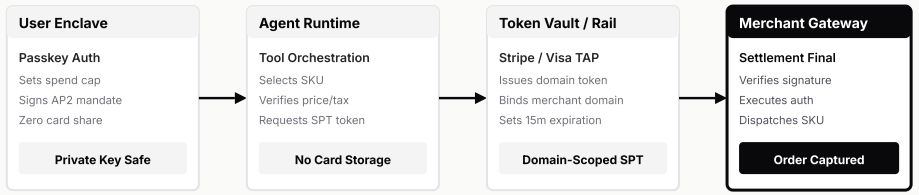


EXHIBIT 4.1 - PAYMENT SETTLEMENT RAILS COMPARISON

Taxonomy and unit economics of card networks, virtual cards, and autonomous rails

Rail Archetype	Processing Cost	Settlement Speed	Dispute Window	Optimal Transaction Type
Tokenized Cards (Visa TAP, SPT)	1.5% - 2.9% + $0.30	T+1 to T+2 Business Days	60 - 90 Days	**Consumer Retail ($15 - $2,000)**
Virtual Cards (Lithic, Highnote)	1.8% - 3.2% + $0.25	T+1 to T+2 Business Days	60 - 90 Days	**Interim Legacy Checkout**
Account-to-Account / FedNow	0.1% - 0.5% (Capped)	Sub-Second Real-Time	Strict / Limited	**B2B Wholesale Procurement**
x402 Stablecoin (USDC Base)	< $0.005 Flat Fee	Sub-Second Finality	Zero / Final	**Sub-Cent API & Data Querying [3]**
Corporate Enclaves (Ramp, Brex)	Interchange Sharing	T+1 Settlement	Internal Admin Rules	**Employee Autonomous Spend**

Settlement Takeaway. Virtual cards bridge legacy web forms but suffer edge-case declines during split shipments [1]. Scalable architectures require native Shared Payment Tokens (SPTs) for consumer goods and x402 micropayments for data feeds [4].

Reference Sources [1] Scott Wessman, Payments Friction Analysis (Oct 2025); [2] Stripe Developer Documentation (2025); [3] nofrontend.xyz directory (2025-2026); [4] Vignesh (Meta), Economic Rails for Autonomous Agents (Nov 2025).

EMPIRICAL EVALUATIONS

Evaluating Agent Accuracy in E-Commerce Workflows

CMU WebArena Academic Benchmarks (Zhou et al.). In academic evaluations conducted by Zhou et al. at Carnegie Mellon University (WebArena, 2024 [1]), autonomous language agents achieved an overall task completion rate of 10.6% to 14.4% on realistic web tasks, compared to 78.2% for human baselines. The benchmark highlights the necessity of justified refusal, where an agent correctly stops execution when product specifications do not match or required shipping parameters are unavailable, rather than hallucinating an erroneous checkout.

VisualWebArena Multimodal Evaluations (Koh et al., ACL 2024). Expanding upon text-based environments, Koh et al. (CMU and Stanford, ACL 2024 [2]) evaluated multimodal agents on complex web interfaces in VisualWebArena. On e-commerce shopping tasks, baseline multimodal agents attained a 20.0% completion rate. Integrating visual tree-search planning increased success to 38.1%, leaving a 61.9% failure rate on visual interface elements such as dropdown selectors, size matrices, and promotional modal overlays.

ACES Benchmark, EComAgentBench, and Tool-Lab Findings. In the ACES shopping evaluation (Allouah et al., Columbia University, Dec 2025 [3]), agents tested across eight product categories displayed pronounced susceptibility to rank positioning and platform badges. A sponsored label reduced selection probability at a fixed rank, while an "Overall Pick" badge increased it. In EComAgentBench (Du et al., Shopee Research, Sept 2026 [4]), testing 662 long-horizon tasks with distributed hidden needs, Claude Opus 4.6 achieved 57.1% accuracy while GPT-5.4 led in individual requirement satisfaction. In Tool-Lab (Wadi & Ma, Sept 2026 [5]), introducing $10 and $50 simulated tool lookup fees caused agents with vague goals to prematurely truncate search, selecting higher unit-cost items.

Behavioral Decision Science - Algorithm Aversion and Control. Experimental studies by Dietvorst, Simmons, and Massey (Wharton, 2015 [6]) document severe algorithm aversion, where consumers avoid algorithms after seeing them err, even when the model outperforms human alternatives. However, follow-up research (Dietvorst et al., Management Science, 2018 [7]) demonstrates that allowing users to modify an agent's forecast or decision by even 2% to 5% substantially eliminates algorithm aversion. Logg, Minson, and Moore (Harvard, 2019 [8]) note baseline algorithm appreciation for objective advice, but this declines when consumers possess strong personal preferences.

FIGURE 5.1 - AGENT CHECKOUT RELIABILITY VS. INTERACTION COMPLEXITY

Scraping brittle HTML interfaces vs. deterministic structured APIs across task steps



EXHIBIT 5.1 - BENCHMARK PERFORMANCE MATRIX

Rigorous empirical evaluations across academic shopping environments

Benchmark	Institution / Scope	Observed Metric	Human Baseline	Key Finding
WebArena	CMU (Zhou et al.) [1]	10.6% - 14.4% Success	**78.2% Human**	Justified refusal is essential
VisualWebArena	CMU / Stanford (ACL 2024) [2]	20.0% - 38.1% Success	**88.7% Human**	Tree search aids visual parsing
ACES Shopping	Columbia (Allouah et al.) [3]	8 Categories x 8 Items	N/A (Synthetic)	Rank & badge bias observed
EComAgentBench	Shopee (Du et al.) [4]	57.1% (Claude Opus 4.6)	N/A (Multi-Turn)	Hidden profile needs neglected
Tool-Lab	Wadi & Ma (2026) [5]	$10 / $50 Tool Fees	N/A (Cost Model)	Cost truncates agent search

Engineering Takeaway. Production deployments must enforce deterministic validation, including sub-50ms stock checks, parameter boundaries, and user confirmation thresholds to prevent catastrophic edge-case drift [7].

Reference Sources [1] Zhou et al., WebArena (CMU); [2] Koh et al., VisualWebArena (ACL 2024); [3] Allouah et al., ACES (Columbia Univ); [4] Du et al., EComAgentBench (Shopee); [5] Tool-Lab; [6] Dietvorst et al. (Wharton, 2015); [7] Dietvorst et al. (Management Science, 2018); [8] Logg et al. (Harvard, 2019).

RISK ARCHITECTURE

Security Frameworks for Machine-Initiated Traffic

The Scale of Automated Web Traffic (Imperva Threat Research). According to the Imperva Bad Bot Report (2024-2025 [1]), automated traffic accounts for 49.6% of total internet traffic. Within this automated volume, 32.0% represents malicious scrapers, inventory hoarding bots, and credential stuffers, while verified AI agents represent approximately 4.8%. Merchants must establish technical mechanisms to distinguish legitimate buyer agents from predatory scrapers without degrading performance or triggering destructive CAPTCHAs that break autonomous flows.

Adversarial Vulnerabilities and Prompt Injection (AgentDojo Benchmark). Security evaluations in the AgentDojo benchmark (Debenedetti et al., 2024-2025 [2]) examine 629 security scenarios across 97 distinct user tasks. Testing indicates that adversarial prompt injections embedded within third-party content can divert agent execution toward unauthorized actions or data exfiltration. The study emphasizes that security requires joint reporting of user task completion without attack alongside success rates under adversarial conditions.

Merchant Catalog Injection Vulnerabilities (Whispers of Wealth Study). The Whispers of Wealth study [3] tested AP2-style agents (Gemini-2.5-Flash) against indirect prompt injections embedded in merchant product descriptions. Across 10 repeated trials, an adversarial description successfully redirected the agent to select the attacker's product in all 10 cases, with the authors judging that six runs violated explicit user constraints. In two runs, synthetic cross-account credential leakage occurred, demonstrating that unverified third-party catalog data poses concrete operational risks.

Cryptographic Request Verification (Visa TAP and RFC 9421). Traditional bot defenses relying on IP reputation, browser fingerprinting, and behavioral mouse tracking fail when inspecting headless API traffic. Visa TAP and RFC 9421 solve this limitation through HTTP Message Signatures. By verifying cryptographic signatures generated by secure enclaves, merchants can validate that an incoming request originates from an authorized user runtime and conforms to an approved spending mandate.

FIGURE 6.1 - INBOUND WEB TRAFFIC CLASSIFICATION MATRIX

Decomposition of global e-commerce traffic based on Imperva Threat Intelligence

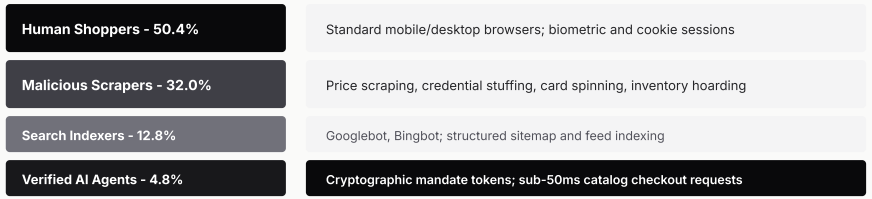


EXHIBIT 6.1 - BOT DEFENSE STRATEGY & THREAT MITIGATION MATRIX

Comparison of traditional WAF heuristics versus cryptographic agent authentication

Threat Vector	Legacy Defense	Agentic Vulnerability	Cryptographic Standard
Prompt Injection	Input sanitization / WAF	Catalog descriptions alter goals [2]	**Deterministic JSON schema bounds**
Inventory Hoarding	CAPTCHA & IP Rate Limits	Blocks valid buyer agents	**15-minute signed cart leases**
Credential Theft	Password & SMS 2FA	Virtual cards lack 3DS automation	**FIDO2 hardware-backed passkeys**
Card Spinning	Velocity rules on PANs	Single-use VCCs trigger false fraud	**Visa TAP RFC 9421 signatures [4]**
Price Scraping	Cloudflare managed challenge	Headless agents cannot solve JS	**x402 pay-per-query API access**

Security Takeaway. Distinguishing legitimate AI buyers from malicious scraping bots requires moving beyond IP reputation toward cryptographic request signing under RFC 9421 and FIDO2 hardware attestation [4].

Reference Sources [1] Imperva Threat Research, Bad Bot Report (2024-2025); [2] Debenedetti et al., AgentDojo (2024-2025); [3] Whispers of Wealth Study (2025); [4] Visa Research Technical Whitepaper (2025); [5] IETF RFC 9421 (2024).

CATALOG OPTIMIZATION

Architecting Product Catalogs for Agent Discovery

Empirical 100,000 Domain Study on Web Readiness (Vatsa Study, Zenodo). In an empirical measurement study of 100,000 domains sampled from the Tranco ranking (Vatsa, 2026, Zenodo record 24121071 [1]), public web readiness for autonomous agents was evaluated across nineteen interface checks. Complete HTML was parsed on 48,442 domains (48.44%). XML sitemap structures were observed on 33,392 domains (33.39%), while structured JSON-LD metadata expansion was supported on 23,436 domains (23.44%). Dedicated agent guidance matching the llms.txt heading profile appeared on 7,156 domains (7.16%). In contrast, declared OpenAPI document schemas were identified on only 225 domains (0.23%), and Model Context Protocol (MCP) initialization handshakes succeeded on only 45 domains (0.045%).

Structured Metadata Specifications (Schema.org JSON-LD). Autonomous agents depend on deterministic metadata rather than subjective advertising copy. Retailers must structure product catalogs using Schema.org JSON-LD, providing explicit fields for price, priceCurrency, availability, itemCondition, hasMerchantReturnPolicy, and shippingDetails. Sites lacking machine-readable schemas force agents to infer specifications through language models, resulting in hallucinated prices and elevated checkout abandonment.

Ursu Position Bias Field Evidence (Chicago Booth). In econometric field research analyzing consumer search on randomized result rankings, R. M. Ursu (University of Chicago Booth School of Business, 2016 [2]) found that higher search rank strongly drives initial clicks and consideration. However, conditional on an item receiving a click, position rank exhibited no detectable direct effect on eventual booking conversion. For agentic commerce, this indicates that appearing in an agent's consideration set is required for discovery, but final conversion is determined by deterministic product attributes (price, delivery time, return terms).

Agent Engine Optimization (AEO) Core Ranking Factors. As consumer product discovery transitions from traditional search engines to conversational agents, search engine optimization (SEO) evolves into Agent Engine Optimization (AEO). Key ranking factors in agent synthesis include structured metadata completeness, inventory query latency (<50ms p95), verified review sentiment, and programmatic return policies. Merchants meeting these requirements secure higher placement in agent-generated comparison sets.

FIGURE 7.1 - AEO DISCOVERY WEIGHTS VS. TRADITIONAL SEO

Algorithmic ranking factors prioritized by LLM purchasing agents during product selection

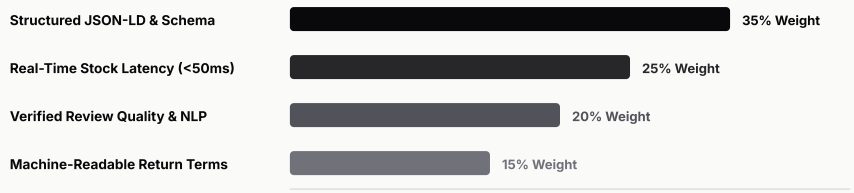


EXHIBIT 7.1 - MERCHANT CATALOG READINESS CHECKLIST

Adoption rates observed across 100,000 domains sampled from Tranco ranking (Zenodo record 24121071) [1]

Technical Requirement	Specification Standard	100k Web Study [1]	Operational Status
Complete HTML Parsed	Server-Rendered Markup	**48.44% (48,442)**	**Basic Baseline**
XML Sitemap Structure	Sitemaps Protocol XML	**33.39% (33,392)**	**Content Discovery**
Structured Metadata	Schema.org JSON-LD Expansion	**23.44% (23,436)**	**High Impact for AEO**
Dedicated Agent Guidance	llms.txt Heading Profile	**7.16% (7,156)**	Developing Standard
Public Machine APIs	OpenAPI Document Schema	**0.23% (225)**	Major Enterprise Gap
Model Context Protocol	MCP Initialize Handshake	**0.045% (45)**	Early Developer Pilot

Readiness Takeaway. While 48.44% of domains return parseable HTML and 23.44% provide structured JSON-LD, programmatic machine interfaces remain exceptionally scarce, with OpenAPI schemas on 0.23% of domains and MCP server endpoints on 0.045% [1].

Reference Sources [1] V. R. Vatsa, "Web Access and Agent Interfaces - A 100,000 Domain Study", Tranco Sample, Zenodo record 24121071 (2026); [2] R. M. Ursu, Chicago Booth (2016); [3] Schema.org Working Group; [4] Shopify Engineering (2025).

MARKET DEFENSIBILITY

Strategic Positioning Across the Agentic Commerce Ecosystem

The Reallocation of Search Advertising Budgets. The transition to autonomous commerce challenges traditional search advertising models. When consumers delegate product discovery to conversational assistants, traditional search result pages and pay-per-click sponsored links ($60B+ annual market) are bypassed in favor of direct synthesis. Retailers must reallocate search acquisition budgets toward structured data endpoints, direct API integration, and agent discovery placement.

Consumer Intent Aggregation Moats (OpenAI, Google, Apple). Front-end conversational platforms possess substantial distribution advantages in intent capture. By operating the direct user interface and managing contextual memory, these platforms capture the initial point of consumer demand. Their strategic challenge centers on executing accurate checkouts without assuming merchant fulfillment or inventory liabilities.

Infrastructure Defensibility of Payment Processors (Stripe, Adyen). While conversational frontends control user interaction, payment infrastructure providers (Stripe, Adyen) maintain durable competitive moats. Processors control merchant acquiring relationships, fraud prevention models, payment rail licenses, and complex tax compliance across global jurisdictions. Replicating this regulatory and operational infrastructure presents high barriers to entry for software aggregators.

Retailer and Marketplace Defensibility (Amazon, Shopify). E-commerce incumbents defend market share through physical logistics, proprietary inventory, and established customer trust. Amazon's fulfillment infrastructure and Prime delivery speeds create defensible moats that pure software agents cannot bypass. Retailers utilizing Shopify or commercetools can maintain direct customer relationships by publishing structured API endpoints, preventing intermediaries from commoditizing their brand equity.

FIGURE 8.1 - VALUE CHAIN POWER DISTRIBUTION

Relative defensibility and margin capture potential across transaction tiers

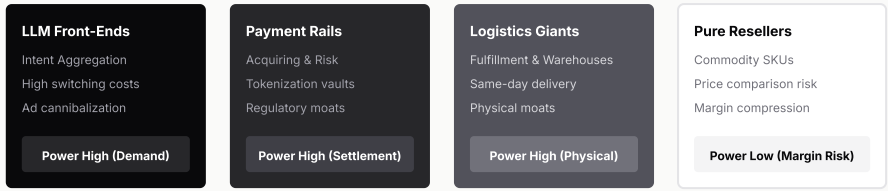


EXHIBIT 8.1 - PLATFORM COMPETITIVE MOATS AND STRATEGIC CAPABILITIES

Analysis of defensive capabilities and architectural advantage across ecosystem layers

Stakeholder Group	Primary Asset Moat	Vulnerability	Strategic Imperative
Front-End AI Assistants	User Context & Prompts	Lack merchant trust & rails	**Partner with Stripe/Adyen [1]**
Payment Processors	Acquiring & KYC Licenses	Displaced if x402 bypasses cards	**Deploy Shared Payment Tokens**
Card Networks	Global Acceptance & Rules	Interchange margin pressure	**Embed Visa TAP / RFC 9421**
Retail Marketplaces	Warehouses & Fulfillment	Search ad revenue erosion	**Launch proprietary shopping agents**
Independent Brands	Proprietary Product IP	Price commoditization risk	**Expose structured JSON-LD feeds**

Strategic Takeaway. Value migrates toward the two ends of the transaction stack, specifically front-end intent aggregators holding user context, and physical logistics networks fulfilling verified deliveries. Middle-tier resellers without proprietary data face severe margin compression.

Reference Sources Morgan Stanley Equity Research (2025); Goldman Sachs Global Payments Analysis (2025); CB Insights Big Tech Strategy Report (2025-2026).

EXECUTION PLAN

An 18-Month Implementation Roadmap for Retail Brands

Phase 1 (Months 1 to 6) - Catalog Modernization and Discovery. The initial phase focuses on exposing structured, machine-readable catalog endpoints. Organizations must implement Schema.org JSON-LD across all SKUs, deploy /.well-known/ucp discovery manifests, and host llms.txt files. Engineering priorities center on optimizing inventory query response times to sub-50ms latencies and establishing automated sitemap feeds for verified search crawlers.

Phase 2 (Months 6 to 12) - Delegated Checkout and Token Integration. In the second phase, retailers integrate delegated checkout protocols, including Stripe ACP and Google UCP endpoints. Payment systems are upgraded to support Shared Payment Tokens (SPTs) and network tokenization (Visa TAP, Mastercard Verifiable Intent). Checkout endpoints must enforce RFC 7231 Idempotency-Key headers to eliminate duplicate transactions during network disconnections.

Phase 3 (Months 12 to 18) - Autonomous Replenishment and Dynamic Pricing. Advanced implementations extend agent capabilities to automated replenishment and machine-to-machine procurement. Retailers deploy AP2 closed mandate validation, integrate x402 micro-settlement for automated catalog access, and establish dynamic pricing algorithms calibrated for algorithmic buyers. Systems enforce multi-tier spending caps and explicit human confirmation thresholds.

Legal and Regulatory Compliance (Christoph Busch Analysis). Legal scholarship by Christoph Busch ("Consumer Law for AI Agents", German Law Journal, 2025-2026 [1]) emphasizes that automated contracts are legally enforceable under US E-SIGN (15 USC 7001(h)) and Washington UETA (RCW 1.80.130), provided an agent's action is legally attributed to the principal. Washington UETA RCW 1.80.090 provides conditional relief for transmission errors only when the consumer has no opportunity to correct the error and provides prompt notice. In the European Union, AI Act Article 50 (effective August 2, 2026) mandates clear disclosure of automated interactions, while consumer protection rules preserve 14-day statutory return rights regardless of agent mediation.

FIGURE 9.1 - 18-MONTH IMPLEMENTATION TIMELINE

Phased enterprise execution across catalogs, token checkout, and autonomous restock

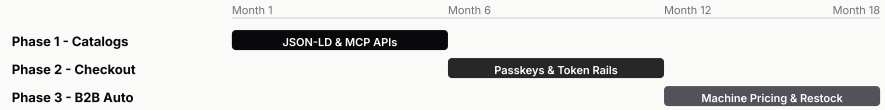


EXHIBIT 9.1 - ENTERPRISE RESOURCE ALLOCATION AND KPI SCORECARD

Milestones, budget distribution, and performance metrics across deployment phases

Phase	Budget Share	Core Technical Milestone	Target Success KPI
Phase 1 - Catalogs	40%	< 50ms stock query latency	99.5% Accuracy
Phase 2 - Checkout	35%	< 2% token decline rate	Zero PAN storage
Phase 3 - B2B Auto	25%	8.2% conversion lift on agents	< 1% return rate

Legal and Regulatory Governance Priorities. Executive committees must review automated agent delegation under US UETA Section 14 and EU AI Act transparency rules to ensure that cryptographic mandates establish legally enforceable contracts [1].

Reference Sources [1] Christoph Busch, "Consumer Law for AI Agents", German Law Journal (Cambridge / Oxford); [2] McKinsey & Company (2025); [3] Bain & Company Technology Practice (2025); [4] Gartner IT Symposium Executive Findings (2025-2026).